Lecture 2
Describing and Visualizing Distributions

Review: Data, Observations and Variables

What are **data**?

Data are a collection of observations or measurements – the values recorded on one or more variables. Data may be structured (fixed rows and columns) or unstructured (free text, images, audio).

What is an **observation**?

A single unit, individual, or case on which measurements are recorded – one store, one skull, one penguin.

What is a **variable**?

A characteristic that is measured or recorded on every observation, and whose value can differ from one observation to the next.

In a data table: each row is an observation and each column is a variable.

Review: Types of Variables

For each scenario: what is the variable being recorded, and what type of variable is it?

1. A researcher interested in pinniped morphology measures the whisker length of 9 harbor seals to the nearest tenth of a millimeter.

Variable: whisker length (mm) – quantitative, continuous

2. For each of 40 animals admitted to a marine mammal rehabilitation center, the attending veterinarian records the species: harbor seal, California sea lion, or northern elephant seal.

Variable: species – qualitative, nominal

3. At each of 15 haul-out sites along the coast, a field team counts the number of pups present.

Variable: number of pups at a site – quantitative, discrete

4. Every admitted animal is also given a body condition score of poor, fair, good, or excellent by the attending veterinarian.

Variable: body condition score – qualitative, ordinal

Where We Left Off

Last class we said:

A **statistic** is a numerical characteristic of data used to describe a property of a **population**: a collection of persons, objects, or things.

That definition leans on the word **population** – but we never said what one actually is.

And we can rarely measure an entire population. So where do the numbers in a statistic actually come from, and what can they tell us about the population we never fully see?

Up next: populations, samples, and sampling.

Sampling and Data

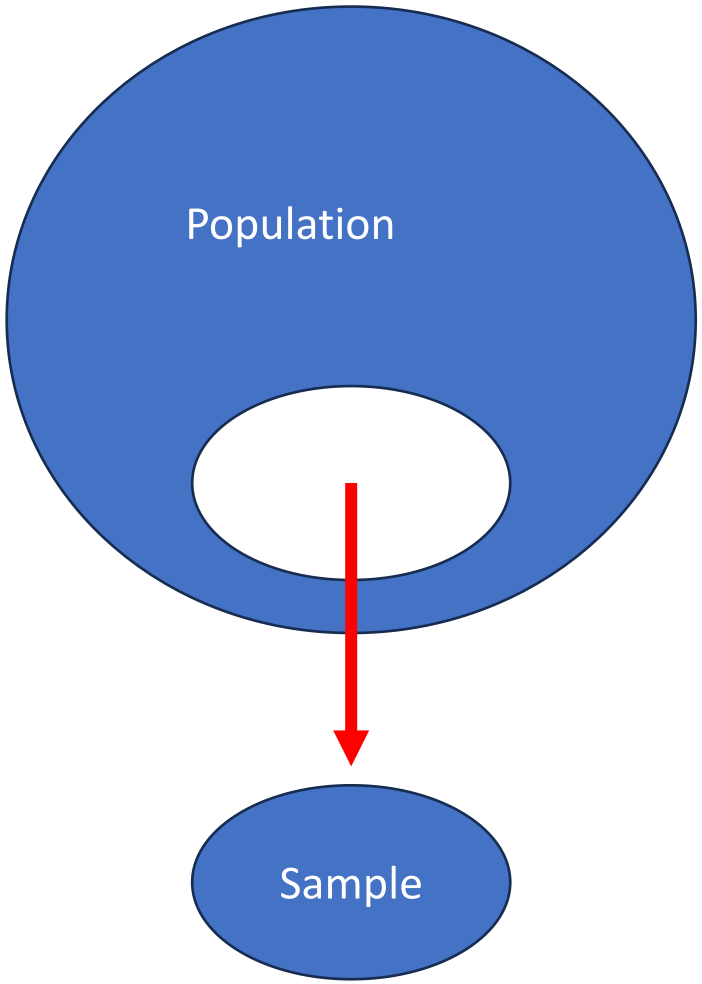
Statistics is generally concerned with studying properties of a **population** – the collection of all possible persons, events, or objects of interest

e.g the set of all possible observations – observed + unobserved

- Populations can be *real* and *finite* (e.g. all employees at a company) or *hypothetical* and *unlimited* (e.g. every hand that could ever be dealt in poker – a process we can repeat indefinitely)

A **Sample** is a subset of the population that we actually observe – the observed observations

- The idea of **Sampling** is to select a portion or individuals or objects that are *representative* of the population
- By studying the sample we can gain insights about the population



A Note on notation:

- Since statistical science is a mathematical discipline, we have a notation that we commonly use:

Fun fact: Leonhard Euler introduced sigma notation in the 18th century. Euler began using the Greek capital letter sigma, Σ, to denote summation, apparently because S is the first letter of *summa* (Latin for "sum").

Example: Penguins at Torgersen Island

The population is a set of N observations
$\{x_1, x_2, x_3, \ldots x_N\}$

The sample is a set of n observations
$\{x_1, x_7, x_8, x_{14}, x_{19}\}$

Population: $N = 20$

Observation	Bill Length (mm)	Bill Depth (mm)	Flipper Length (mm)	Body Mass (g)	Sex
▸ **1**	**39.1**	**18.7**	**181**	**3750**	**male**
2	39.5	17.4	186	3800	female
3	40.3	18.0	195	3250	female
4	NA	NA	NA	NA	NA
5	36.7	19.3	193	3450	female
6	39.3	20.6	190	3650	male
▸ **7**	**38.9**	**17.8**	**181**	**3625**	**female**
▸ **8**	**39.2**	**19.6**	**195**	**4675**	**male**
9	34.1	18.1	193	3475	NA
10	42.0	20.2	190	4250	NA
11	37.8	17.1	186	3300	NA
12	37.8	17.3	180	3700	NA
13	41.1	17.6	182	3200	female
▸ **14**	**38.6**	**21.2**	**191**	**3800**	**male**
15	34.6	21.1	198	4400	male
16	36.6	17.8	185	3700	female
17	38.7	19.0	195	3450	female
18	42.5	20.7	197	4500	male
▸ **19**	**34.4**	**18.4**	**184**	**3325**	**female**
20	46.0	21.5	194	4200	male

Source: Palmer Station Antarctica LTER. Gorman, K. B., Williams, T. D. and Fraser, W. R. (2014). PLoS ONE 9(3):e90081. R package palmerpenguins (Horst, Hill and Gorman, 2020). All twenty Adélie penguins at Torgersen Island, 2007; observation 4 was banded but not measured, and sex was not determined for observations 9–12.

Sample: $n = 5$

Observation	Bill Length (mm)	Bill Depth (mm)	Flipper Length (mm)	Body Mass (g)	Sex
1	**39.1**	**18.7**	**181**	**3750**	**male**
7	**38.9**	**17.8**	**181**	**3625**	**female**
8	**39.2**	**19.6**	**195**	**4675**	**male**
14	**38.6**	**21.2**	**191**	**3800**	**male**
19	**34.4**	**18.4**	**184**	**3325**	**female**

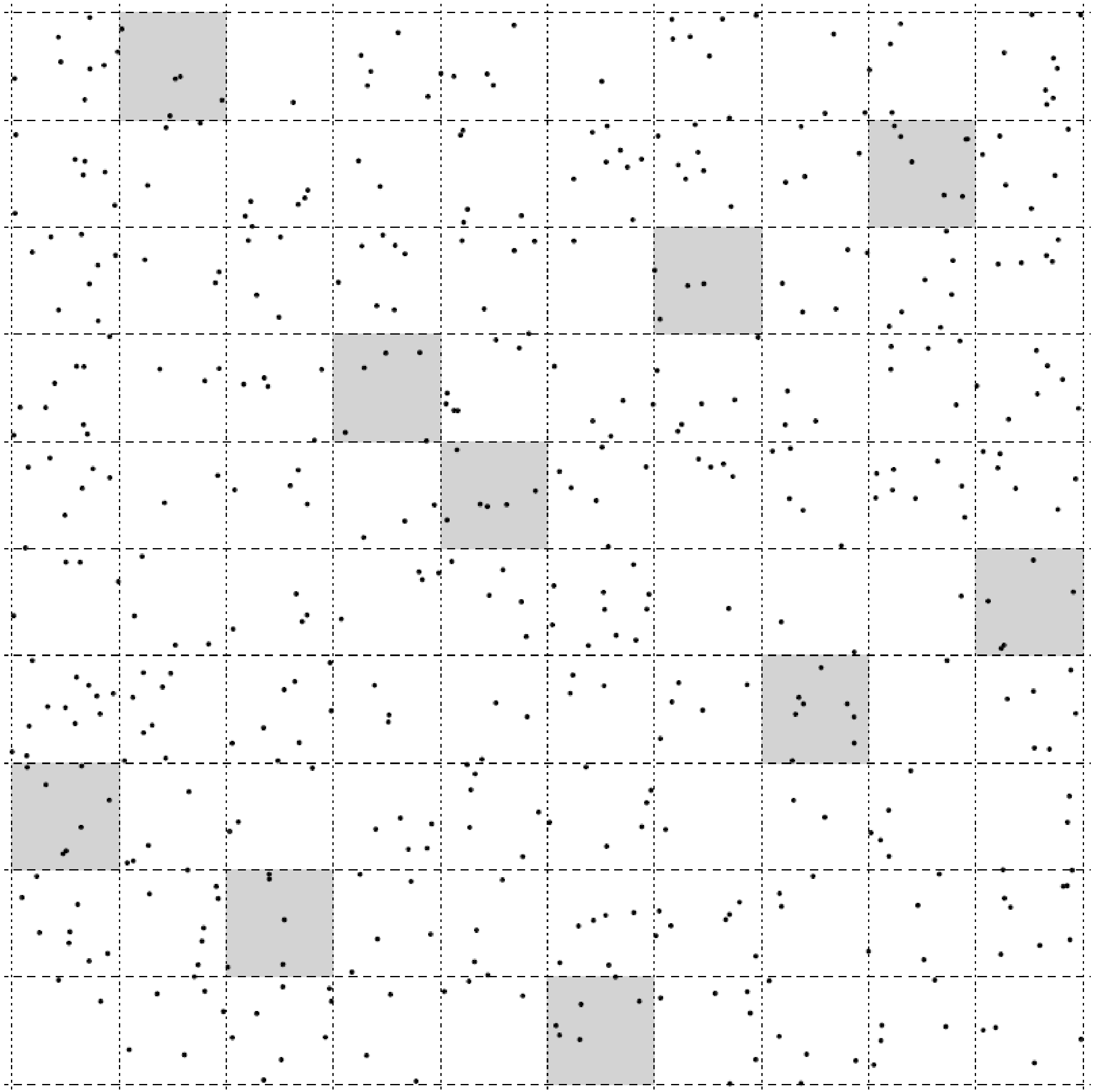
Example 2

Consider a rectangular-shaped piece of land that has been divided into 100 smaller rectangular units, each containing something of interest (e.g., trees, burrows, archaeological artifacts).

A subset of 10 of those smaller units was selected and the number of objects in each of these units was counted.

Population Size: $N = 100$

Sample Size: $n = 10$



Why is statistics so valuable?

- As we said at the start: most of the time we can’t measure everyone or every unit in the population, and so we must limit our measurements to a sample.

- A central task in statistics is **estimation** – the process of inferring an unknown quantity about a population using a set of sample data.

- The tools for estimation let us approximate almost anything about a population **using only a well-chosen sample**.

And they tell us how far off we are likely to be – which is what separates an estimate from a guess.

Where we can go with estimates

- With **estimates** we can
  - Assess differences among groups and relationships between variables.
  - Describe populations. Examples of estimates include averages, proportions, measures of variation, and measures of relationship.
  - Then we can ask and answer questions or formally, test and evaluate hypotheses.

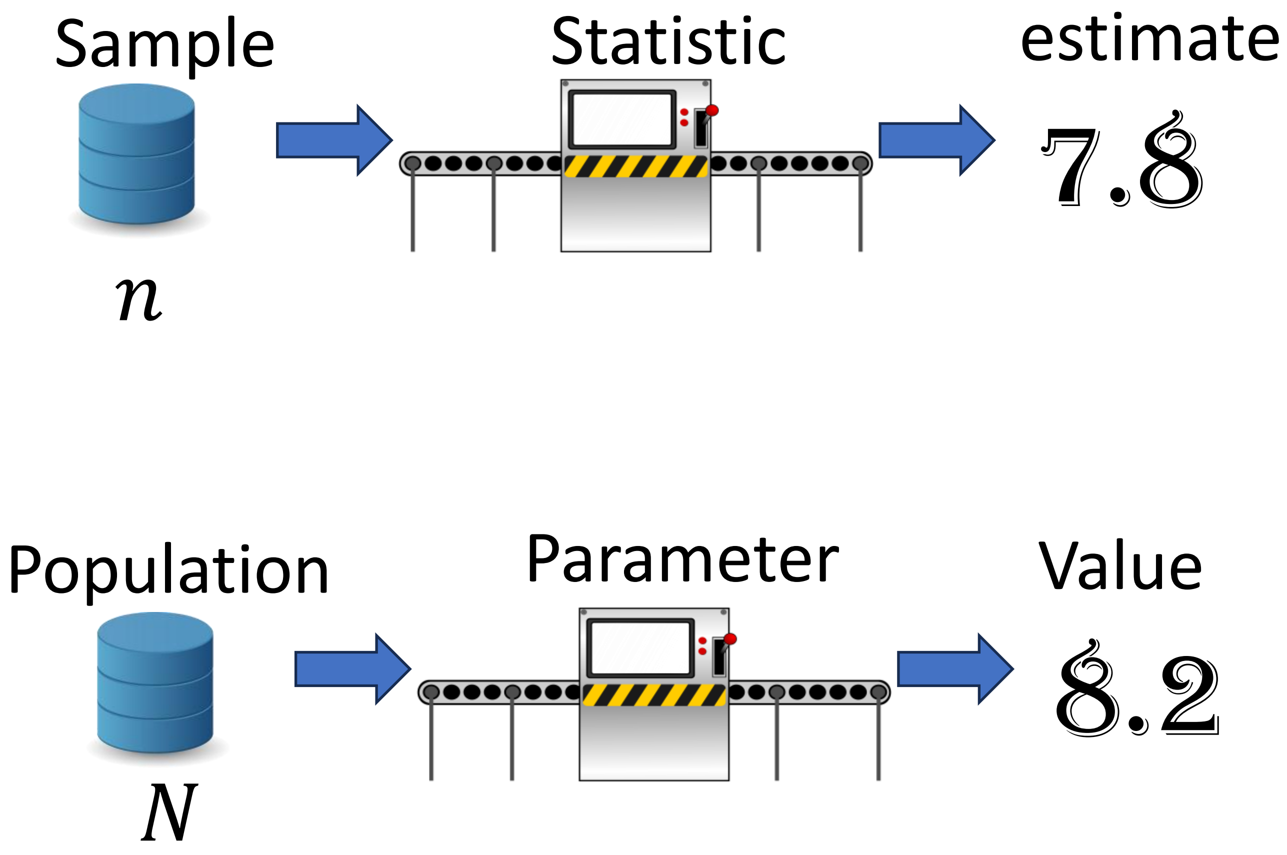
Statistics Vs Parameters

A **statistic** is a numerical characteristic of a **sample** that estimates a population parameter.

A **parameter** is a numerical characteristic of a **population** that can be estimated by a statistic.

Put another way…

A **statistic** is a function of the observations of in a sample while a **parameter** is a function of all observations in the population.



Mean and Proportion

A **proportion** describes the fraction of a whole that represent some property or category. Usually, it is expressed a percentage.

Notation:

$\hat{p}$ - denotes the sample proportion

p - denotes the population proportion

The arithmetic **mean** is the center of a set of data (we often use the words mean and average interchangeably)

Notation:

$\bar{x}$ - denotes the mean of a sample

μ – denotes the mean of a population (i.e the population parameter)

Parameter

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$p = \frac{\text{Number of objects in category}}{N}$$

Statistic

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\hat{p} = \frac{\text{Number of objects in category}}{n}$$

Example: Gallup Poll

- On April 20, 2010, one of the worst environmental disasters took place in the Gulf of Mexico when the Deepwater Horizon offshore oil rig exploded.
- In response to the spill, many activists called for an end to offshore drilling. Almost nine months later, turbulence in the Middle East pushed the price of oil to an all-time high.
- In March 2011, Gallup conducted a survey and found that 60% of Americans favored offshore drilling as a means to reduce U.S. dependence on foreign oil.
- The poll interviewed 1,021 adults aged 18 and older in the continental U.S., selected by random digit dialing.
- **What is the population, and what parameter is being estimated?**

  All adults 18 and older in the continental U.S.; the parameter is p, the true proportion of them who favor offshore drilling.

- **What is the sample, and what is the statistic?**

  The 1,021 adults interviewed ($n = 1{,}021$); the statistic is $\hat{p} = 0.60$.

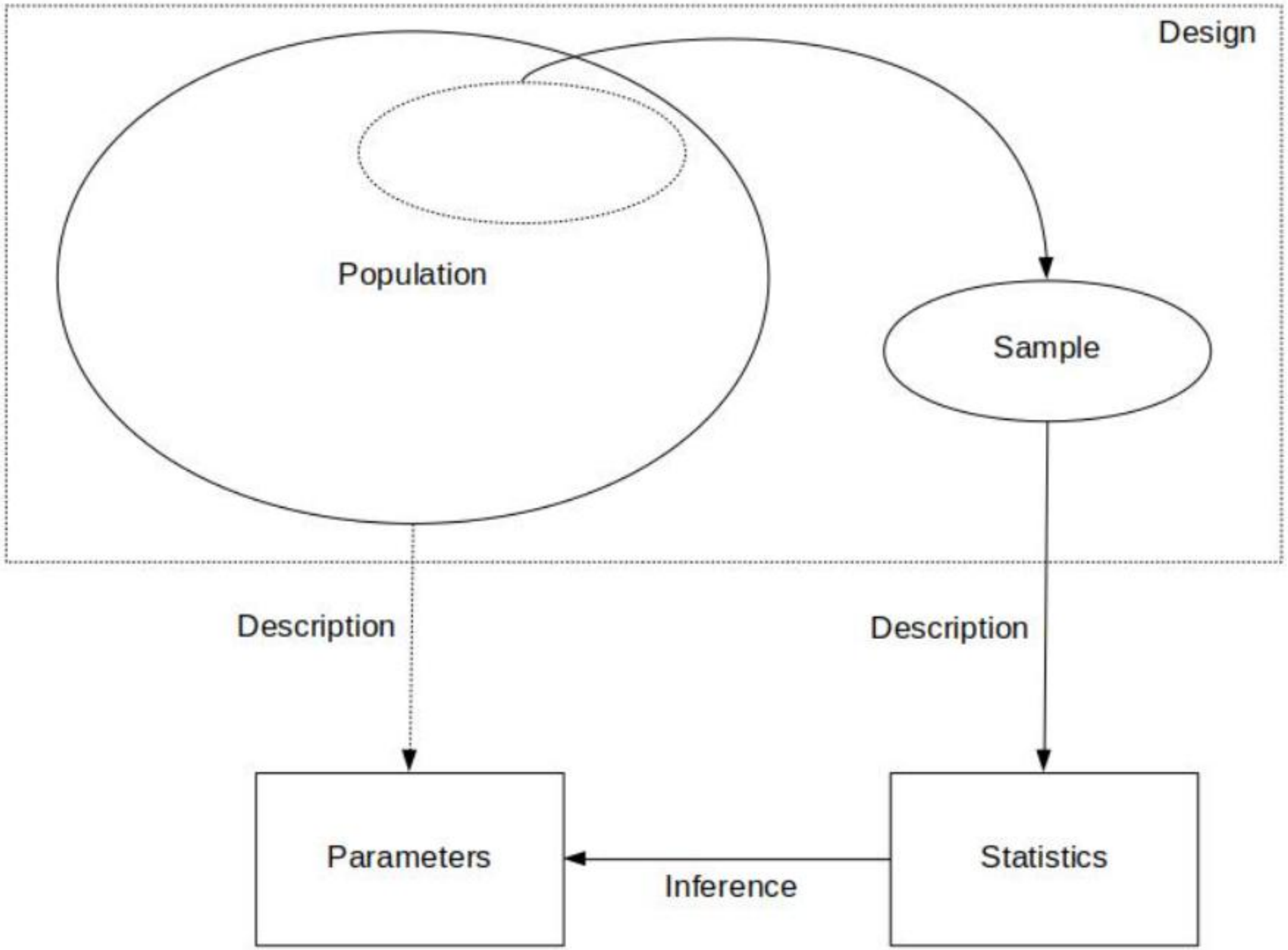
Descriptive Vs. Inferential Statistics

1. **Design** – The process/method in which we plan to collect data to answer our statistical question

2. **Descriptive Statistics –** refers to describing the observations in a sample using statistics or a population using parameters
    1. - collection, organization, summarization and visualization of data

3. **Inferential Statistics** (or statistical inference) – refers to using a sample (usually a statistic) to answer a question about a population (such as estimating the value of a parameter)

    - estimation, hypothesis testing, determining relationships among variables, prediction


Design
Population
Sample
Description
Description
Parameters
Inference
Statistics

Warm Up

In elections, television networks use exit polling (interviewing voters after they leave the voting booth) to declare the winner well before all votes are counted. In the 2010 California gubernatorial election between candidates Jerry Brown (D) and Meg Whitman (R), an exit poll projected Brown to be the winner early into election night. Specifically, the network responsible for the poll interviewed 3,889 voters at the booth and determined that 53.1% favored the democratic candidate.

What is the statistical question?

Who will win the California race for governor?

What is the population? What is the population size N?

All Californians who actually voted – not everyone eligible. N is unknown that night.

What is the sample? What is the sample size n?

The 3,889 voters interviewed as they left the booth; n = 3,889.

What is the statistic being calculated?

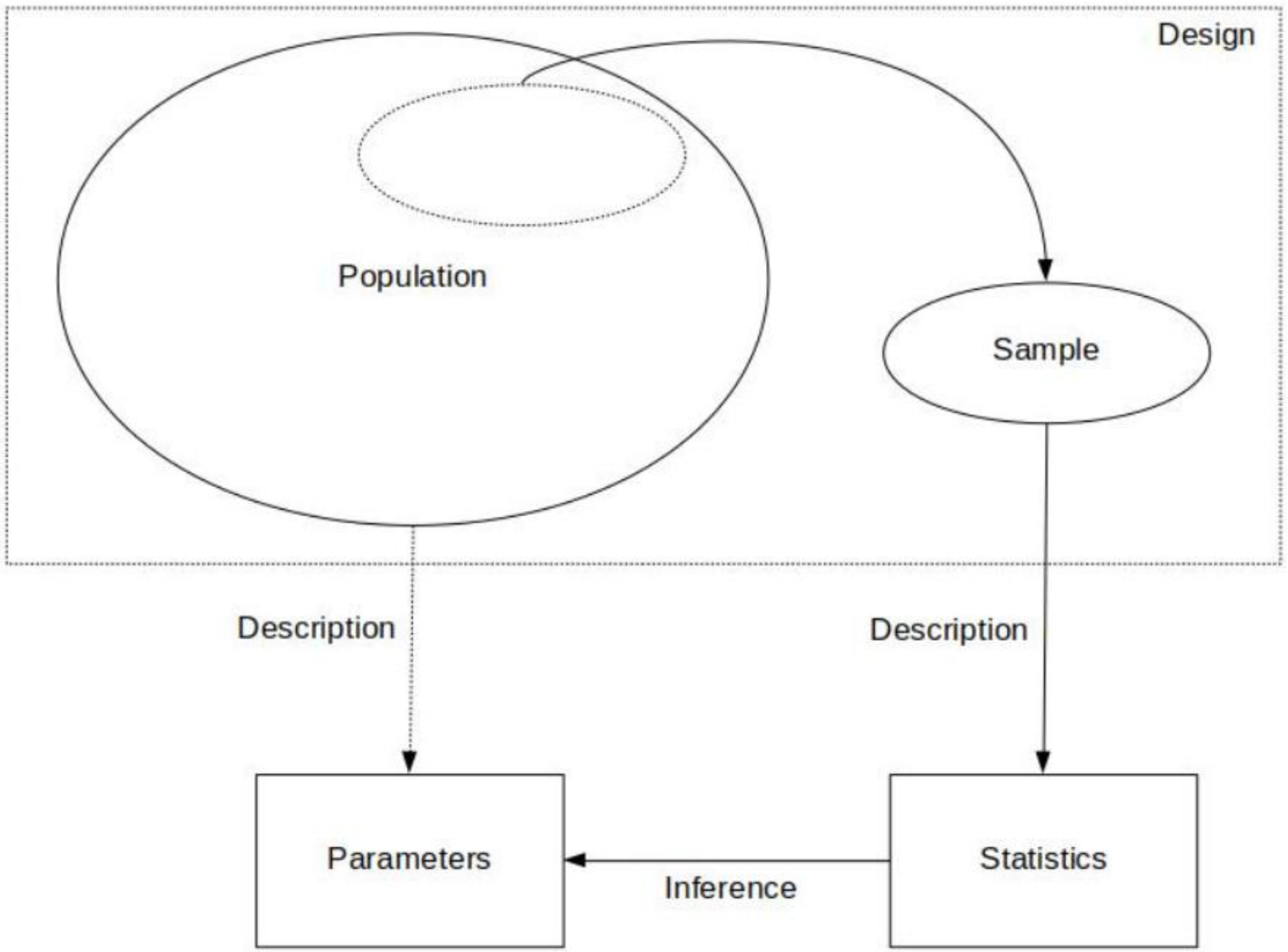
$\hat{p}$ = 0.531, the proportion of those 3,889 who chose Brown.

Recap:

1. **Design** – the goal/statistical question we want to answer and how we plan to obtain our data

2. **Description** – a preliminary exploration and summary of the data

3. **Inference** – using statistics to make decisions or predictions about the data



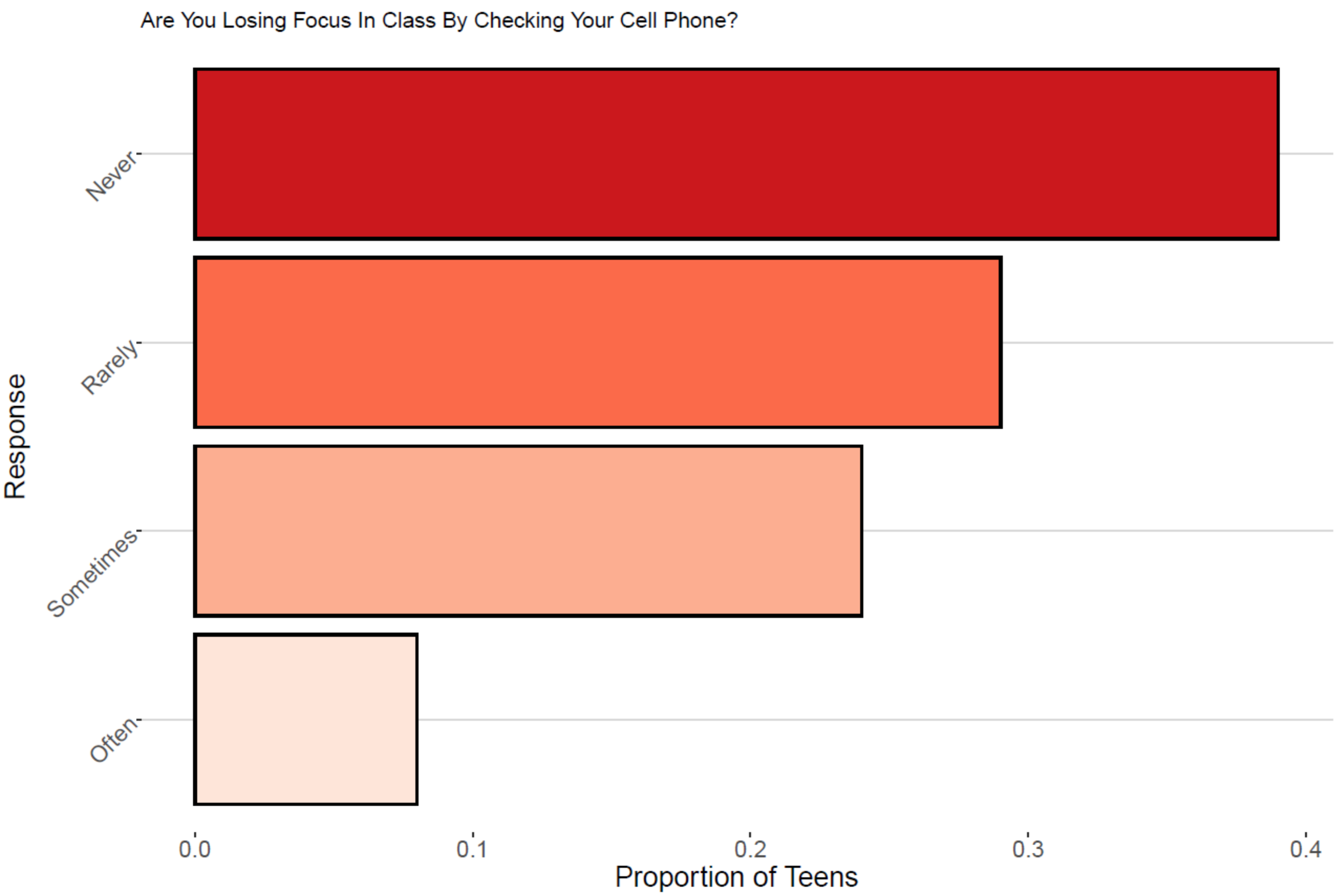
Descriptive statistics

- Methods for summarizing, visualizing and characterizing data
- <u>The data can be from samples OR populations</u>
- Goal: simplify the data without distorting or losing information
- Summaries are easier to understand

Ex. Are teens distracted by their cell phones? A study conducted by the Pew Research center surveyed 743 U.S teenagers ages 13-17 to understand how cell phones in class impacted their ability to concentrate. Each student was asked to rate the impact of cell phone use on their concentration based on a Likert scale

Student	Age	Response
1	13	Never
2	13	Sometimes
3	15	Never
4	17	Often
⋮	⋮	⋮
743	16	Rarely

Possible Responses: Never, Rarely, Sometimes, Often


Are You Losing Focus In Class By Checking Your Cell Phone?
Never
Rarely
Sometimes
Often
Response
0.0
0.1
0.2
0.3
0.4
Proportion of Teens

Features of Distributions

- A **distribution** of a variable gives (a) the values that occur and (b) how often each value occurs

<u>Features of A Distribution</u>

Categorical Variables:

- **Modal category** – the category with the highest frequency

Quantitative Variables:

- **Shape** – do observations cluster into certain areas, are values spread out or more densely packed together?

- **Center** – where is the middle point on the distribution, where does a typical value fall?

- **Variability** – how tightly to observations cluster around the center of the distribution

Displaying Distributions: Frequency Tables

- **Frequency table –** a table listing the distinct values of variable together with the number of observations of each value

- **Frequency** – the number of times a value occurs

- **Relative Frequency** – the proportion of observations that assume a given value

$$\mathrm{RF} = \frac{f}{n},$$

- **Cumulative Relative Frequency** - the proportion of observations equal to or less than a given value (more on this later)

Ex. Are teens distracted by their cell phones?

Response	Frequency	Relative Frequency	Cumulative Relative Frequency
Never	289	0.39	0.39
Rarely	216	0.29	0.68
Sometimes	178	0.24	0.92
Often	60	0.08	1.00

Ex. Mendels Pea Plants - In one of Gregor Mendel's classic studies, he bred 8023 pea plants and observed the color of the pea pods.

Pea Color	Frequency	Relative Frequency
Yellow	6022	0.751
Green	2001	0.249